



ENGINEERED SOLUTIONS
FOR A CLEANER, SAFER TOMORROW

RELIABLE SOLUTIONS.
REAL PROGRESS.

DOSING
INJECTION
BLENDING
TRANSFER
CONTROL

RELIABLE
FLEXIBLE
CUSTOMIZABLE

CENTRIFUGAL DOSING SKID SYSTEM LNS-DZ-CEN

Multi-Industry Chemical Dosing & Inline Blending System

PEOPLE
INDUSTRY
CLEANER
TOMORROW



FUELS &
TERMINALS



CHEMICAL &
PETROCHEMICAL



INDUSTRIAL
PROCESS



WATER
TREATMENT



UTILITIES
& OTHERS

SOLUTIONS
FOR A SAFER TOMORROW

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PRODUCT OVERVIEW & APPLICATION SCOPE

1. PRODUCT DESCRIPTION

The **LINESIG LNS-DZ-CEN** is a modular centrifugal-pump dosing skid designed for medium- to high-flow chemical dosing, additive injection, inline blending and controlled transfer duties across multiple industries. The system can be configured as a single-train package or as a multi-pump skid with independent metering and control on each discharge line.

The design is suitable for continuous-duty applications where a stable, non-pulsating flow is preferred. Typical services include treatment chemicals, process additives, fuels and hydrocarbon additives, ethanol or oxygenate blending, water-treatment chemicals, utility chemicals and general industrial liquid injection. Final pump hydraulics, wetted materials, seals and instrumentation are selected to suit the actual fluid properties and process conditions.

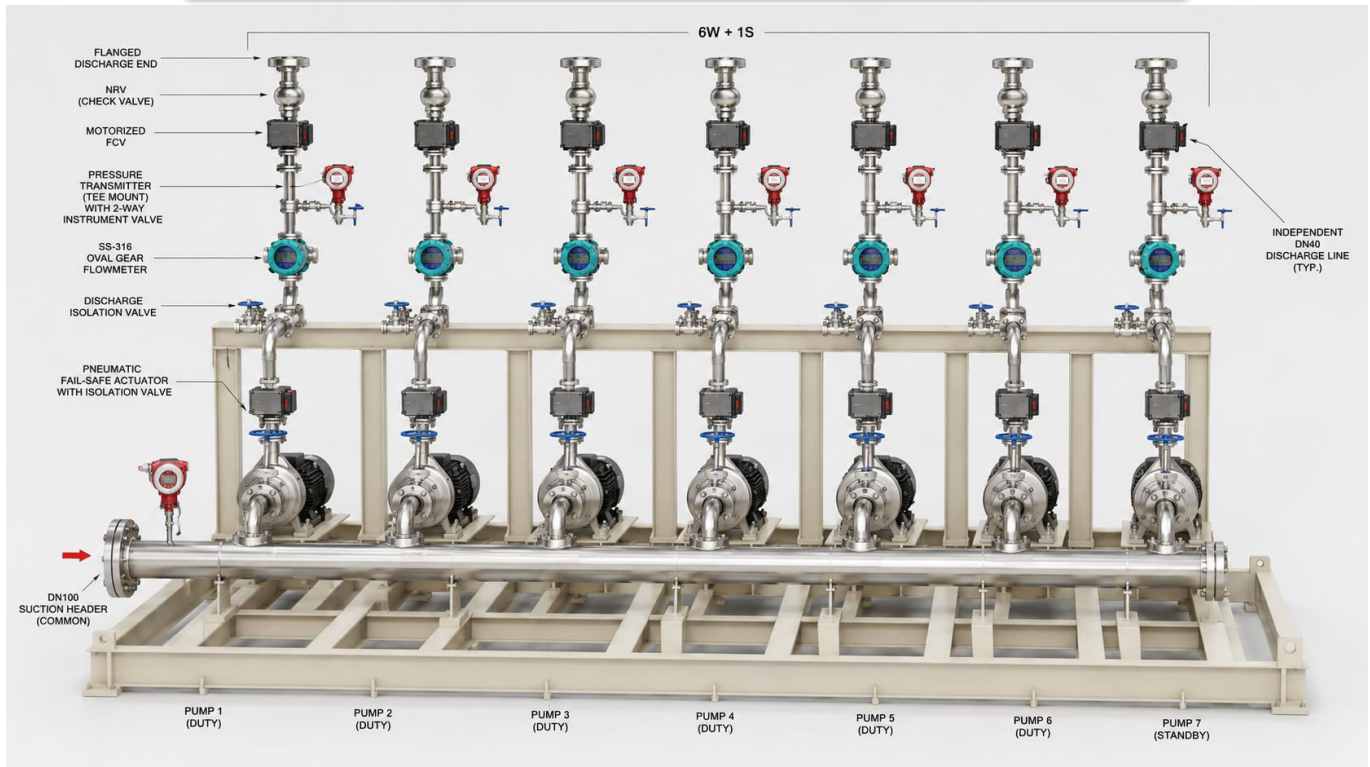
2. INDUSTRIES

- Oil & gas, refinery, terminal and tank-farm facilities
- Petrochemical and chemical processing plants
- Fuel terminals, loading gantries and inline blending systems
- Water and wastewater treatment
- Power generation and utility chemical systems
- Food, process and general manufacturing
- Industrial depots, bulk storage and transfer facilities

3. TYPICAL MEDIA

Media Group	Typical Examples	Engineering Note
Hydrocarbon / Fuel Additives	Ethanol, methanol, oxygenates, lubricity improvers, dyes, markers, additives	Seal, motor hazardous-area rating and material compatibility to be confirmed.
Water Treatment Chemicals	Coagulants, antiscalants, neutralization chemicals, conditioning liquids	Viscosity, corrosivity and crystallization tendency to be reviewed.
Industrial Chemicals	Solvents, process chemicals, cleaning liquids, selected acids/alkalis	Not all aggressive chemicals are suitable for centrifugal duty; pump type may change.
Utilities	Cooling-water chemicals, boiler-water chemicals, wash-water additives	Continuous low-to-medium pressure duty.

REFERENCE SKID CONFIGURATION (6W + 1S)



4. REFERENCE CONFIGURATION

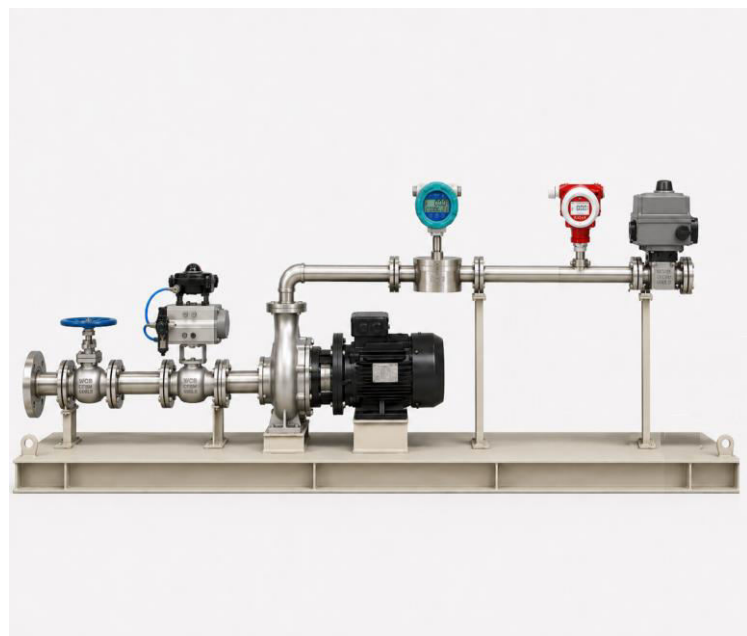
Parameter	Reference Basis
Pump Arrangement (ref)	6 Working + 1 Standby centrifugal pumps (can be customized)
Common Suction Header	DN100 typical
Individual Pump Suction	Project specific; typically, DN50 class
Individual Discharge	DN40 typical
Per-Line Metering	Independent flowmeter on each discharge
Per-Line Control	Motorized control valve / FCV
Pressure Monitoring	Pressure transmitter on each line
Isolation / Check Valves	Provided as required for safe isolation and reverse-flow prevention
Skid Construction	Modular structural frame, outdoor-duty finish

5. SINGLE PUMP TRAIN

The LNS-DZ-CEN can also be supplied as a compact single-pump dosing or injection module. This configuration is suitable for one process line, one product, a local additive package, or a pilot / expansion service. Multiple single trains can later be manifolded or operated independently.

6. TYPICAL SINGLE-TRAIN EQUIPMENT

- Suction isolation valve and optional strainer
- Centrifugal pump with electric motor
- Discharge check valve / non-return valve
- Flowmeter suitable for the fluid and accuracy requirement
- Motorized or pneumatic flow-control valve
- Pressure indicator / pressure transmitter
- Manual isolation valves and drain / vent connections
- Local junction box or compact control panel, when required



TECHNICAL SPECIFICATIONS

7. GENERAL DESIGN DATA

Parameter	Typical / Available Basis
Service	Chemical dosing, additive injection, inline blending and controlled liquid transfer
Flow Range	Project specific; typical medium/high-flow service from approximately 5 to 100 m ³ /h per train
Differential Pressure	Project specific; typically low to moderate pressure, subject to pump curve
Operating Temperature	Project specific according to liquid and material selection
Viscosity	Best suited to low- and moderate-viscosity liquids within centrifugal-pump capability
Pump Type	End-suction / close-coupled / process centrifugal pump as required
Drive	Electric motor; VFD optional
Duty Philosophy	Single, duty/standby, or multiple working + standby
Skid MOC	Carbon steel with protective coating; stainless steel optional
Wetted MOC	SS316/316L typical; other metals or non-metallics by service
Connections	ASME / EN flanged or project-specific
Installation	Indoor or outdoor; hazardous-area version available subject to project classification

8. HYDRAULIC SELECTION

Final centrifugal pump selection shall be based on the actual design flow, differential pressure, static head, piping losses, fluid density, viscosity, vapor pressure, NPSH available, temperature and allowable operating range. The operating point should be selected near the pump best-efficiency region where practical.

PROCESS CONTROL & METERING

9. CONTROL PHILOSOPHY

Each dosing line may operate independently. The required chemical or additive flow is measured and controlled to a fixed setpoint, ratio setpoint or remote command from a PLC, DCS, batch controller or terminal automation system.

Control Function	Typical Implementation
Flow Measurement	Coriolis, oval gear / PD, magnetic, ultrasonic or other technology according to fluid and accuracy
Flow Control	Motorized FCV, pneumatic FCV or VFD speed control
Pressure Monitoring	Local gauge and/or pressure transmitter
Temperature Monitoring	RTD / temperature transmitter on common header or process line where required
Ratio Blending	Chemical flow setpoint derived from main process flow
Batch Operation	Preset quantity with totalization and automatic stop
Remote Operation	PLC/DCS/SCADA command with permissive and feedback
Alarm Functions	Low flow, high/low pressure, motor trip, valve fail, communication fail

10. METERING SELECTION

For high-accuracy hydrocarbon and additive applications, Coriolis or positive-displacement flowmeters may be preferred. For conductive aqueous chemicals, electromagnetic flowmeters may be suitable. Final meter technology shall be selected according to conductivity, viscosity, density, required accuracy, flow turndown and pressure loss.

PIPING, VALVES & MATERIALS

11. PIPING ARRANGEMENT

- Common suction manifold sized to minimize pressure loss and prevent pump starvation.
- Independent pump suction branches with isolation valves.
- Independent discharge lines to each injection / blending point.
- Check valves provided where reverse flow from process headers is possible.
- Drain, vent and flushing points incorporated where required.
- Supports arranged to minimize nozzle loads and vibration.
- Flexible connections or expansion allowances added where thermal movement requires.

12. MATERIAL SELECTION

Service	Typical Wetted Material Options
Hydrocarbon / Fuel Additive	SS316L, carbon steel where compatible, suitable elastomers
Alcohol / Oxygenates	SS316L commonly selected; seal/elastomer compatibility to be verified
Corrosive Water Chemicals	SS316L, duplex, alloy, PP/PVDF or lined components depending on chemistry
Solvents	SS316L or compatible alloy; seal compatibility essential
General Water-Based Chemical	SS316L or suitable non-metallic materials

13. SEALS

Mechanical-seal arrangement shall be selected according to toxicity, flammability, vapor pressure, lubricity and leakage tolerance. Single, double or sealless pump options may be considered where service severity requires.

AUTOMATION, SAFETY & ELECTRICAL

14. PLC / HMI INTEGRATION

- Local PLC/HMI package or remote DCS/SCADA integration
- Auto / manual operating modes
- Pump duty selection and standby changeover
- Flow-ratio or batch control
- Valve position monitoring and command
- Totalization and batch reports
- Alarm/event history where PLC/HMI supplied
- Remote permissive from loading gantry, process line or terminal system

15. SAFETY FEATURES

Protection / Interlock	Typical Function
Low Suction Pressure / Level	Prevent dry running or loss of prime
High Discharge Pressure	Protect pump and downstream piping
Low / No Flow	Detect pump, valve or blockage problem
Motor Overload / Trip	Electrical equipment protection
Emergency Stop	Local skid E-stop and optional remote ESD interface
Leak / Spill Detection	Optional tray sensor or hydrocarbon detector
Earthing / Bonding	Provided as required for hazardous or hydrocarbon service

16. HAZARDOUS AREA

For flammable hydrocarbon, alcohol or solvent service, motors, instruments, junction boxes and electrical components shall be selected for the project hazardous-area classification. The final Ex/ATEX/IECEx requirement is project specific and is not fixed by the skid model alone.

APPLICATION CONFIGURATIONS & SELECTION

17. TYPICAL SYSTEM CONFIGURATIONS

Configuration	Recommended Use
1 x 100%	Non-critical single duty or local injection
2 x 100% (1W + 1S)	Critical duty with full standby redundancy
2W + 1S	Two independent dosing lines with common standby
4W + 1S	Multi-line dosing package
6W + 1S	Terminal / gantry multi-line reference configuration
N Working + 1 Standby	Customized modular systems

18. WHEN TO SELECT LNS-DZ-CEN

- Medium to high liquid flow is required.
- Continuous, smooth, non-pulsating flow is preferred.
- Injection pressure is within practical centrifugal-pump capability.
- Multiple parallel lines require independent flow measurement and control.
- Inline blending or transfer duty is more important than very-low-flow metering accuracy.
- The fluid is compatible with centrifugal pump hydraulics and does not require a positive-displacement metering pump.
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19. WHEN TO SELECT ANOTHER LINESIG DOSING FAMILY

Duty Requirement	Recommended Family
Very accurate low-flow chemical feed	LNS-DZ-PDD diaphragm metering skid
High-pressure injection	LNS-DZ-PDP piston/plunger skid
Very low-flow precision additive / catalyst dosing	LNS-DZ-PDG precision micro gear skid
Medium/high-flow smooth dosing or inline blending	LNS-DZ-CEN centrifugal skid

ORDERING DATA & STANDARD SCOPE

20. MINIMUM DATA REQUIRED FOR FINAL SIZING

- Chemical / product name and concentration
- Density and viscosity at minimum / normal / maximum temperature
- Vapor pressure and flammability classification where applicable
- Required flow per line and turndown
- Suction pressure / tank level range
- Required discharge pressure and downstream backpressure
- Number of dosing lines and standby philosophy
- Required flowmeter technology / accuracy
- Control mode: manual, local automatic, ratio, batch or remote PLC/DCS
- Material of construction and seal requirements
- Hazardous-area classification
- Power supply and control voltage
- Applicable project standards and inspection requirements

21. TYPICAL STANDARD SCOPE

Item	Typical Supply
A	Skid structural frame and equipment supports
B	Centrifugal pump(s) and motor(s)
C	Common suction manifold
D	Individual suction and discharge piping
E	Isolation and check valves
F	Per-line flowmeter
G	Per-line control valve or VFD control arrangement
H	Pressure instrumentation
I	Local piping drains / vents as applicable
J	Junction box or PLC/HMI panel when specified
K	Internal electrical wiring and cable supports when panel supplied
L	Documentation: datasheet, GA, P&ID, I/O list, operating philosophy and test records as applicable



RELIABLE DOSING. PRECISE CONTROL. MULTI-INDUSTRY FLEXIBILITY.

LINESIG Dosing Skid Systems — Engineered for process reliability, safe operation and scalable integration.